

STTM Extractor

Overview & How It Works

Executive Summary

The STTM Extractor is a Python-based automation accelerator that generates Source-to-Target Mapping (STTM) documents by parsing Informatica PowerCenter mapping XML files. It eliminates the traditionally manual, error-prone process of documenting ETL data flows — reducing effort by up to 95% while improving consistency and quality. The tool is part of a five-component Informatica AI Suite designed to transform how over 200 people work on Informatica-based platforms.

What is STTM?

STTM stands for Source-to-Target Mapping — a critical document in data integration that describes how data flows from source systems to target systems, including all transformation logic, business rules, joins, filters, and lookups applied along the way.

STTMs serve multiple critical functions:

- Migration Planning — Defines what needs to be rebuilt on the target platform during data migration.
- Compliance Documentation — Satisfies regulatory requirements (e.g., BCBS 239) by demonstrating how data flows through systems.
- Impact Analysis — Reveals which downstream targets are affected when a source system changes.
- Developer Onboarding — Helps new team members understand existing data flows without reading thousands of lines of legacy code.
- Data Quality Validation — Provides the expected transformation logic against which actual data can be validated.

What is the STTM Extractor?

The STTM Extractor is one of five integrated tools in the Informatica AI Suite — a platform investment designed to transform delivery across Informatica-based programs. The complete suite includes:

Tool	Description
STTM Extractor	Automatically generates Source-to-Target Mapping documents from ETL metadata.
Lineage Builder	Constructs end-to-end data lineage across ETL layers.

Impact Analyzer	Determines downstream effects of schema or logic changes.
SQL Backfill Generator	Generates SQL scripts for backfill operations.
On-Demand Data Comparator	Compares data between old and new systems for validation.

Individually useful — together, these tools form a delivery transformation engine. This is what a platform bet looks like: one investment, system-wide impact across 200+ team members.

How Does It Work?

Input

The STTM Extractor takes Informatica PowerCenter mapping XML files as input. These XML files are exported from the Informatica repository and contain the full definition of every transformation, port, expression, and connection within a mapping.

- Supported: Informatica mapping XML files (individual or batch)
- Not currently supported: Workflow XML files (can be built on demand)

Processing

The tool uses a Python-based, rule-driven approach to parse and interpret the mapping metadata:

- Parses the XML metadata using Python to extract mapping structures — source tables, target tables, fields, data types, transformations, joins, lookups, filters, and more.
- Analyzes the mappings the same way a human analyst would — interpreting transformation logic, conditions/rules for Lookups, Joiners, and Filters, Source Qualifier overrides, and comments at mapping, table, and field levels.
- Implements deterministic, rule-based automation to interpret transformations. The solution is Python and technology-heavy, not AI-heavy — ensuring accuracy, repeatability, and auditability.
- Leverages GitHub Copilot or similar AI coding assistants to accelerate development while maintaining code quality.

Output

The tool produces a standardized Excel spreadsheet (STTM document) at the mapping level, containing the following fields:

Output Field	Description
Source System Details	Source system name, source table, field names, and data types
Target System Details	Target table, field names, data types, and target instance name
Transformation Flow	Complete ETL transformation logic within the mapping
Calculations	Expressions and calculations used in transformations
Conditions / Rules	Conditions and rules for Lookup, Joiner, and Filter transformations
Source Qualifier Overrides	Any SQL overrides applied at the Source Qualifier level
Comments	Comments captured at mapping, table, and field levels

Version & Subject Area	Version number and subject area metadata
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Architecture

The STTM Extractor is built as a modern, Python-based accelerator designed for modularity, portability, and ease of deployment. Key architectural characteristics include:

- Python-based accelerator — parses Informatica mapping XML files directly without requiring a live repository connection or database access.
- Modular design with deterministic, rule-based parsing logic that can be extended to support additional transformation types and edge cases.
- Development accelerated using GitHub Copilot and AI coding assistants, while ensuring all outputs are accurate and auditable.
- Outputs standardized Excel-based STTM documents aligned with the approved template format.
- Designed to run within the TD Bank network environment to meet security and compliance requirements.
- Lightweight footprint — no dependency on Java, application servers, or dedicated database schemas; runs as a standalone Python application.

Validation & Current Status

The STTM Extractor has undergone rigorous review and testing:

- The STTM mapping document has been reviewed and validated with TD stakeholders (Savio, Sanjeev, Sumathi) and is working as per expectation.
- Tested with 50+ sample mappings across the complexity spectrum — covering simple to highly complex ETL patterns.
- Additional enhancements to the STTM template have been suggested by stakeholders and are being incorporated.
- Business Systems Analysts (BSAs) have already started using and verifying the generated STTM documents as part of the delivery workflow.
- A request to process 250+ SHAW mappings has been received, demonstrating production-scale demand.

Where It Fits — Composable AI Workflow

The STTM Extractor is the first agent in a composable AI workflow that chains specialized agents to create end-to-end intelligent delivery capabilities. The workflow follows this pattern:

Workflow Step	Description
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Extract Metadata	STTM Extractor parses mapping XMLs and extracts structured metadata.
Classify	Metadata Classifier categorizes data elements, sensitivity levels, and business domains.
Generate Mapping	Documentor auto-creates mapping documents, data dictionaries, and business rule summaries.
Build Lineage	Lineage Builder constructs end-to-end data lineage across all ETL layers.
Analyze Impact	Impact Analyzer determines downstream effects for governance and change management.

This composable approach reduces manual dependency, improves consistency across teams, and lets the organization build once and reuse many times. Outputs become standardized artifacts — not one-off personal notes.

Key Benefits

Benefit	Details
Up to 50% Effort Reduction	Automates STTM document creation that traditionally takes hundreds of analyst-hours.
Improved Quality	Deterministic, rule-based parsing produces consistent, error-free outputs every time.
Faster Time to Market	Accelerates migration, modernization, and compliance initiatives.
Scalable	Can process large batches of mappings (50+ tested, 250+ requested) without additional manual effort.
Reusable Platform	Part of a composable suite — not a point solution — that transforms delivery across the portfolio.
Audit-Ready	Standardized outputs support compliance, regulatory reporting, and governance requirements.